

INFO 371: *Voice Assistant Review* *Sentiment Classifier*

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Over the course of two days, you will work with your book club group (and occasionally with help from other groups) to plan, build, and evaluate models for two different applications. You are not graded on building a “good” model for the application, but on having completed all steps and providing reasonable justification for your work.

Assigned Application

All groups are assigned to two applications. You will be graded as an *entire book club group*, but will probably want to break up into subgroups for completion!

Assignments are found at the [INFO371 Book List Sheet](#).

General Instructions

- Create *two* copies of this document, one for each application
- Put the name of the application and group/member names at the top
- In class, in time with the rescheduled activities, complete each section! **Replace** all italicized text with your own responses (you can work in subgroups or with your whole book club group)
- When complete, make a pdf copy of *both* documents and have **one** member of the group upload both documents.

Schedule

- Day 1:
 - [Part I: Review your assigned applications and make general plan](#)
 - [Part II: Write clarification questions](#)
 - Part II: Get clarification answers
 - [Part III: Plan Modeling Approach](#)
 - [Part IV: Start Implementation](#)
- Day 2:
 - Part IV: Finish Implementation
 - [Part V: Model Report](#)
 - Part VI: [Debrief](#) with own group
 - Part VI: Debrief with Proposing Group

Grading

The following provides a general guideline for grades. Grades for the full book club are calculated as the average for the two separate applications (each application is marked as out of 5, the whole thing is worth 10 pts; each “day” has work worth about half of all the points)

- Part I (0.5 pts total)
 - 0.5 pts: You have provided reasonable answers for all components
 - 0.25 Points: You provided responses that were not reasonably justified using concepts from the course or skipped some sections
- Part II (0.5 pts total)
 - 0.5 pts: You wrote relevant questions and got answers
 - 0.25 Points: You wrote partly irrelevant questions and got answers
 - 0 Points: You did not consult with the other group to get answers
- Part III (1 pt total)
 - Full Points: You have provided reasonable answers for all components
 - $\frac{3}{4}$ Points: You provided several responses that were not reasonably justified using concepts from the course
 - $\frac{1}{2}$ Points: You skipped some sections
- Part IV (1 pt total)
 - Full Points: You have linked to runnable, readable code that completes all the objectives
 - $\frac{3}{4}$ Points: You have linked to code that is missing an important section, is almost entirely lacking in formatting (is very hard to read), or contains sections that do not run (but would have easy fixes)
 - $\frac{1}{2}$ Points: You are missing multiple important sections of code or analysis (for instance, missing model selection for *two* approaches or missing evaluation or preprocessing)
- Part V (1 pt total)
 - Full Points: You have provided reasonable answers for all components
 - $\frac{3}{4}$ Points: You provided several responses that were not reasonably justified using concepts from the course
 - $\frac{1}{2}$ Points: You skipped some sections
- Part VI
 - Full Points: You have provided reasonable answers for all components for your own and the other group
 - $\frac{1}{2}$ Points: You have provided reasonable answers for all components for your own but are missing responses from the other group

Part I: General Planning

Read through your applications and check out the data (make sure you can load it and do some preliminary data exploration to visualize some data, check for the types of values in the data, missing data, the kinds of transformations that might be required). Replace the italic text in the following questions. You may revise this after you check in with the proposing group.

Using the Data Set

- What is the target variable of interest?
 - *The target variable is whether a review expresses positive or negative sentiment about the voice assistant.*
 - *feedback*
 - *The feedback column already contains binary labels (1 = positive, 0 = negative), so it can mostly be used as it appears. We should still check for missing values and confirm that it is stored as a numeric binary variable before training the model.*
- What are the features that you think will be most useful for prediction?
 - **verified_reviews**
This is the most important feature because it contains the text of the review. The words in the review can show whether the customer feels positive or negative.
 - **rating**
The star rating may help predict sentiment. Lower ratings are more likely to be negative reviews, while higher ratings are usually positive.
 - **variation**
This column shows the product model. Some models may receive different types of feedback, so it may help the model learn patterns.
- What are the features that you **won't** use for prediction?
 - **date**
The review date probably does not affect whether a review is positive or negative, so it is not very useful for prediction.
- Are there important non-linearities or interaction effects to consider?
 - *There may be some interaction effects between words in the reviews. For example, phrases like "not good" or "very helpful" may have different meanings than the individual words alone. Because of this, using features like bigrams could help capture these interactions. However, since the dataset is not very large, starting with simple word counts is a reasonable approach.*
- What pre-processing or feature engineering will be required?
 - *Remove missing review text to avoid errors during model training.*
 - *Convert the review text into numerical features using CountVectorizer, which transforms each review into word-count vectors.*

- *Remove common stop words (such as “the”, “is”, “and”) so the model focuses more on meaningful words that indicate sentiment.*

Formal Model Evaluation

- What is/are the most important formal evaluation metric(s) to report?
 - *Metric 1: Recall (for negative reviews)*
 - *Measures how well the model correctly identifies negative reviews. This is important because missing negative feedback could hide problems with the product.*
 - *Metric 2: Precision (for negative reviews)*
 - *Measures how many reviews predicted as negative are actually negative. This helps ensure the model is not incorrectly labeling too many positive reviews as negative.*
- What are further analyses that would be useful?
 - *One useful analysis would be to examine whether predicted sentiment differs across device variations. Some device models may receive more negative feedback than others, which could indicate product issues or design flaws.*
 - *To check this, we will calculate the average predicted sentiment for each device variation and visualize the results using a Plotly bar chart. This will allow us to see whether certain device versions receive more negative predictions than others.*

Other Model Considerations

- What are the explainability needs?
 - *Explainability is somewhat important because developers may want to understand which words lead to positive or negative predictions. Models like logistic regression allow us to see which words have strong positive or negative weights.*
- What are the latency needs?
 - *Latency is not very critical because sentiment classification does not require instant responses. Training may take a few seconds, and prediction for a single review should be very fast.*
- What are the evolution needs?
 - *The model may need to be retrained periodically as new reviews are collected. This allows the model to adapt to new words or changing user opinions over time.*
- What are the fairness needs
 - *Fairness is less critical for this application because the model predicts sentiment from product reviews rather than personal attributes. However, it is still important to ensure the model does not systematically misclassify certain types of reviews.*

Success Criteria

For the following, you should provide numerical values for any metrics that you would check, as well as non-numerical things that you would check to know if your model is a “good choice”.

Write at least 3 things to check off

- Our model would be “successful” if...
 - a. *The model provides useful insights from the review data, such as identifying device variations that may receive more negative feedback.*
 - b. *The model achieves high precision for negative reviews, so positive reviews are not incorrectly flagged as negative.*
 - c. *The model provides useful insights from the review data, such as identifying device variations that may receive more negative feedback.*

Part II: Proposal Q&A

After you have had time to skim through the proposal and have a quick look at the data, you should write 3 questions to ask members of the original proposing group. These questions should be designed to give you a better sense of how to do the analysis. They can be clarification about the application scenario or the available data, the priorities of the proposing group, opinions about potential metrics, or checking in about further analysis of the model that might be useful.

- Question 1: *Should the model treat violent language, rude language, and dismissive/condescending language as equally important, or is one type more important to catch than the others?*
 - Answer: *The model should treat all types of offensive language similarly. Methods such as TF-IDF or CountVectorizer convert words into numerical features, allowing the model to analyze different types of language patterns in a similar way.*
- Question 2: *Is it worse for the model to miss offensive speech or to incorrectly flag neutral speech as offensive?*
 - Answer: *In this scenario, incorrectly flagging neutral speech (false positives) could be more problematic because it may create unnecessary concern or lead to incorrect moderation decisions.*
- Question 3: *Should the model focus only on clear offensive words, or should it also try to detect subtle tone, such as sarcasm, passive-aggressive phrasing, or condescension?*
 - Answer: *Ideally, the model should also detect subtle tone such as sarcasm, passive-aggressive phrasing, or condescension. However, due to time and complexity limitations in this assignment, focusing on clearly offensive words is a more practical approach.*

Part III: Modelling Plan

Consider the following big picture modeling approaches: k-nearest neighbors, linear models with or without regularization (lasso or ridge), logistic models with or without regularization (lasso or ridge), decision trees, random forests, or an approach that doesn't use formal modeling approaches (such as using summary statistics). You can also consider SVM, naive bayes, or other approaches if you like.

You should choose a first and second choice for a big picture modeling approach based on what you think will be most useful. For your analysis, you will need to select a single final model. For both your first and second choice, you will need to compare different settings (hyperparameters, feature sets, preprocessing choices) and then compare the best model from the first approach and the second approach to choose your ultimate final choice to evaluate. For each approach, you must consider multiple “**contenders**” that vary in *at least* one aspect (for instance, different settings for a single hyperparameter or different feature combinations) but you are encouraged in many cases to consider additional combinations. For instance, you might try out logistic regression with four different lasso regularization values *and* a random forest model with 3 different maximum depth values.

First Choice Approach

- Approach:
 - *Logistic Regression with text features from CountVectorizer*
- Reason for choice:
 - *Logistic regression is commonly used for text classification and works well with high-dimensional word features. It is also simple, fast, and relatively easy to interpret.*
- Specific Considerations
 - Overfitting Risk
 - *There is some risk of overfitting because text data can produce many features. However, logistic regression generally handles high-dimensional data well.*
 - Explainability
 - *Logistic regression is somewhat interpretable because the model coefficients show which words contribute positively or negatively to the prediction.*
 - Latency
 - *Training and prediction should be fast since logistic regression is computationally efficient.*
 - Evolution
 - *The model would likely need to be retrained periodically as new reviews are collected.*

- Contenders to Check Out
 - *Hyperparameter: Regularization strength (C)*
 - *Reason for Consideration: Different values of C control how strongly the model penalizes large coefficients, which can affect model performance and overfitting.*

Second Choice Approach

- Approach:
 - *Random Forest with text features from CountVectorizer*
- Reason for choice:
 - *Random forest can capture more complex patterns in the data by combining multiple decision trees. It may detect relationships between words that linear models like logistic regression cannot.*
- Specific Considerations
 - Overfitting Risk
 - *Random forests reduce overfitting compared to a single decision tree by averaging predictions from many trees. However, with many text features the model could still become complex.*
 - Explainability
 - *Random forests are less interpretable than logistic regression because the predictions come from many trees, making it harder to directly understand which words influence the prediction.*
 - Latency
 - *Training may take longer than logistic regression because the model builds many decision trees. However, prediction should still be reasonably fast.*
 - Evolution
 - *If new reviews are added, the model would need to be retrained with the updated dataset to incorporate new information.*
- Contenders to Check Out
 - *Hyperparameter: Number of trees*
 - *Reason for Consideration: The number of trees controls how many decision trees are included in the forest. Testing different values may improve performance while balancing model complexity and training time.*

Part IV: Execute the Analysis

You should conduct an analysis and implement the code to build and evaluate a model that addresses the task. You can work in different notebooks, but should ultimately produce a single Colab notebook that includes all steps to clean, preprocess, build models, and evaluate them.

Provide a link to the Colab notebook (**accessible** to anyone with a UW account):

- [!\[\]\(e43e4b1050bbff3ab0579ed9f5c72143_img.jpg\) *Voice Assistant Review Sentiment Classifier.ipynb*](#)

Your notebook should fulfill the following:

- It should be **runnable** from beginning to end without producing errors (assuming that the data file is loaded)
- It should be **readable** – include headers and at least minimal explanation in markdown that make it clear what different sections or code chunks are *for* (you don't need to say what they *do* unless there's something particularly weird; say the goal of each section)
- It should include necessary code for **data clearing** or **preprocessing** and code required to **explore** the data to know what preprocessing is necessary (for instance, check for missing data, linearity, etc.)
- It should include all code necessary for **model selection** your final model choice (you should look at two modeling approaches and multiple variations for both); you **must use holdout or cross-validation**, you can do it in a loop or use shortcuts, you can do validation for both approaches in the same code cells or in separate cells
- It should include all code necessary for formally **evaluating** your final model choice
- It should include at least one **visualization** created with **plotly express**.

Suggested (not required) headers in your document include:

- Data Exploration & Preparation
- Model Selection
- Model Evaluation
- Further Analysis

Part V: Model Report for Voice Assistant Review Sentiment Classifier

We believe we were not fully successful at meeting the requirements of this application.

We identified the following success criteria:

- *precision for negative reviews is at least 0.85*
- *recall for negative reviews is at least 0.80*
- *the model provides useful variation-level insights*

Final Approach

The final model selected was Logistic Regression with text features from CountVectorizer. During model selection we tested Logistic Regression with different regularization strengths and Random Forest with different numbers of trees. Logistic Regression with $C = 10$ achieved the highest recall on the validation set and was selected as the final model.

Model Evaluation

Formal Metrics

- *Accuracy: 0.946*
- *Precision: 0.758*
- *Recall: 0.490*

These results indicate that the model correctly identifies many reviews overall, but it does not detect enough negative reviews. Because the recall and precision values are lower than the required thresholds, the model does not fully satisfy the application requirements.

Further Analysis

- *The dataset is highly imbalanced, with far more positive reviews than negative reviews. This imbalance likely makes it more difficult for the model to correctly identify negative reviews, which may explain the lower recall.*

Other Considerations

- *Logistic regression offers some explainability because the learned coefficients show which words contribute to positive or negative predictions. The model is also computationally efficient, making predictions very quickly.*

Production Considerations

Expected performance

- *Real-world performance may be slightly lower than the test results because new reviews may contain different wording or topics that were not present in the training data.*

Future Directions

Potential Improvements

- *One improvement would be to try TF-IDF features instead of simple word counts, which could help the model give more importance to informative words and reduce the impact of very common words.*
- *Another improvement would be to test additional models such as Naive Bayes or Support Vector Machines, which are often effective for text classification tasks.*

Part VI: Debrief

You should complete each of the following with your own group first and then meet with members of the proposing group and exchange thoughts.

Check Requirements

- Our group: *Our group believes we were not fully successful, because the model did not reach the required precision and recall thresholds for detecting negative reviews.*
- Proposing group: *The proposing group also agrees with us.*

Comfort Using the Model

- Our Group: *We would not feel completely comfortable using the current model in a real customer support system because it still misses a lot of negative reviews. However, with some improvements and better handling of the data imbalance, a stronger model could still be useful for helping support teams find unhappy customers more quickly.*
- Proposing Group: *The proposing group also agrees with us.*

Benefits

- Our Group: *Customer support teams would benefit from this application because it could help them quickly find negative reviews and respond to customers who may be having problems. Product managers and engineers could also use this information to identify common issues with their devices. Overall, it could help companies respond to feedback faster and improve their products.*
- Proposing Group: *The proposing group also agrees with us.*

Harms

- Our Group: *Even if the model performs well, there could still be harm. Companies may rely too heavily on automated predictions and pay less attention to the full context of customer feedback. This could lead to some customer concerns being misunderstood or overlooked.*
- Proposing Group: *The proposing group also agrees with us.*

Connect to an issue or application in *your own book*

- *In Artificial Intelligence for Fashion, AI models are used to analyze data and help companies make decisions, such as predicting trends or recommending products. Similarly, the salmon disease detector uses machine learning to analyze fish characteristics and predict whether a fish is diseased. Both show how AI can use data patterns to support decision-making in different industries.*

Connect to an issue or application in *proposing group's book*

In 2-4 sentences, summarize an issue, concept, or phenomenon about this application that was in the proposing group's book.

- *In their book, they discuss how voice assistants like Alexa are often designed with feminine personalities, which can reinforce gender stereotypes. This relates to our application because analyzing user reviews could help companies understand how people feel about these devices and their design.*

Talked with TA, the group are not presented mostly, discuss the last question with proposal group